DAY 3 — Exploring Pandas for Data Analysis

Day 3 focused on **Pandas**, which is the most widely used Python library for handling dataset tables. Pandas introduces **DataFrames** — a tabular way of storing data, just like Excel sheets.

🔹 **What I Learned About Pandas:**

I began by loading datasets using:

- pd.read_csv()
- pd.read_excel()
- pd.read_json()

Once the dataset was loaded, I explored it using:

- df.head() to view first rows
- df.info() to view datatypes
- df.describe() for statistical summary
- df.shape for size
- df.columns to see column names

🔹 **Data Cleaning**

A major part of Day 3 was learning how to clean and prepare data, which is the most important step in Data Science. I practiced:

- Removing duplicate rows
- Dealing with missing values (drop or fill)
- Converting datatypes
- Renaming columns
- Handling null values
- Cleaning inconsistent entries

This step is crucial because **clean data = accurate results**.

🔹 **Filtering & Conditional Selection**

I learned to filter data based on conditions such as:

- Price > certain value
- Volume < threshold
- Date-based filtering

Using .loc[] and .iloc[], I selected specific rows and columns efficiently.

🔹 Grouping & Aggregation

I practised:

- groupby() for grouping data
- sum, min, max, count
- Multi-index grouping

These skills are used heavily in time series and financial analysis.

🔹 Merging Multiple Datasets

I also learned how to merge tables using:

- merge()
- concat()
- join()

This is extremely useful when working with different sources of data.

Key Takeaways from Day 3:

- Pandas makes data cleaning simple and powerful
- DataFrames are ideal for structured datasets
- Filtering, grouping, and merging are essential skills
- Pandas plays a major role in preparing data for machine learning